

ASCII MASTER FLOW – PANEL 1/12: Biotechnology information rail

BIOLOGICAL INPUT -> gene / cell / tissue / enzyme / organism

DNA --TRANSCRIPTION--> RNA --TRANSLATION--> PROTEIN

INTERVENTIONS -> rDNA / PCR / culture / diagnostics / bioprocess

OUTPUT -> knowledge / service / regulated product

RULE -> biotechnology is broader than genetic engineering

ASCII MASTER FLOW – PANEL 2/12: Recombinant DNA assembly chain

TARGET GENE -> restriction endonuclease cut

VECTOR -> plasmid / bacteriophage / Ti-plasmid carrier

DNA LIGASE -> recombinant construct

HOST TRANSFORMATION -> marker selection -> expression

TRAP -> vector, insert, host and product are different

ASCII MASTER FLOW – PANEL 3/12: PCR distinction matrix

PCR CORE -> denaturation -> annealing -> extension

THERMOSTABLE POLYMERASE -> repeated DNA copying

REVERSE-TRANSCRIPTION PCR -> RNA first converted to cDNA

REAL-TIME PCR -> fluorescence measured during amplification

RULE -> template conversion and detection timing are separate axes

ASCII MASTER FLOW – PANEL 4/12: Bioprocess scale rail

STRAIN / CELL LINE + MEDIUM -> upstream preparation
BIOREACTOR -> pH / temperature / oxygen / sterility control
HARVEST -> downstream recovery and purification
QUALITY CONTROL -> identity / purity / reproducibility
LAB SUCCESS != PILOT != COMMERCIAL MANUFACTURE

ASCII MASTER FLOW – PANEL 5/12: Culture and potency map

PLANT EXPLANT -> sterile medium -> callus / shoots / roots

TOTIPOTENCY -> suitable cell can regenerate whole plant

EMBRYONIC STEM CELL -> pluripotent

ADULT STEM CELL -> usually multipotent / tissue-restricted

iPSC -> somatic cell reprogrammed to pluripotency

ASCII MASTER FLOW – PANEL 6/12: Gene therapy and diagnostics twin rail

THERAPY -> add / replace / silence / edit disease-relevant sequence

DELIVERY -> engineered viral vector or lipid nanoparticle

DIAGNOSTIC -> sample -> extraction -> marker detection -> interpretation

SOMATIC != GERMLINE; MARKER != CLINICAL OUTCOME

RULE -> mechanism, delivery, validation and status all matter

ASCII MASTER FLOW – PANEL 7/12: Biotechnology institution router

DBT / MINISTRY OF S&T -> policy, missions and research support
BIRAC -> industry-academia translation and enterprise support
ICMR / CSIR / ICAR -> domain-specific research systems
CDSCO -> drugs and vaccines; GEAC -> GMO environmental release
RULE -> funder, performer, interface and regulator are distinct

ASCII MASTER FLOW – PANEL 8/12: BioE3 six-sector wheel

- 1 BIO-BASED CHEMICALS AND ENZYMES
- 2 FUNCTIONAL FOODS AND SMART PROTEINS
- 3 PRECISION BIOTHERAPEUTICS
- 4 CLIMATE-RESILIENT AGRICULTURE
- 5 CARBON CAPTURE AND ITS UTILISATION
- 6 FUTURISTIC MARINE AND SPACE RESEARCH

ASCII MASTER FLOW – PANEL 9/12: Bio-RIDE and GenomeIndia timeline

28 FEB 2024 -> DBT announces 10,000 GenomeIndia sequences complete

24 AUG 2024 -> Cabinet approves BioE3

19 SEP 2024 -> Cabinet approves Bio-RIDE

GENOME DATA -> reference and research potential plus safeguards

APPROVAL / COMPLETION VERB -> preserve exact object and status

ASCII MASTER FLOW – PANEL 10/12: Research-to-impact funnel

DISCOVERY -> proof of concept

BIRAC / INCUBATOR -> translation and enterprise support

BIOFOUNDRY / PILOT -> reproducibility and process optimisation

REGULATION / QUALITY / MANUFACTURE -> adoption gateway

SOCIAL BENEFIT -> affordability, access and fit still required

MUST REMEMBER: Biotechnology uses genes, cells, tissues, enzymes, microbes and...

ASCII MASTER FLOW – PANEL 11/12: PYQ applications matrix

BIOPHARMA -> rDNA + culture + bioprocess + quality + access

MICROBES / FUEL -> digestion + fermentation + biomass conversion

OBJECTIVE CONCEPTS -> barcode / biofilter / Wolbachia / air DNA

CLEAN TECH -> bounded contribution to wider energy transition

2026 -> genetic medicine and GenomeIndia; no provisional key asserted

CLOSE DISTINCTION: Biotechnology is not only genetic engineering, PCR is not sequencing,...

ASCII MASTER FLOW – PANEL 12/12: Biotechnology answer spine

DEFINE -> platform and biological information level

EXPLAIN -> mechanism from laboratory to bioprocess

MAP -> DBT / BIRAC / research bodies / regulators

APPLY -> health / agriculture / industry / environment / energy

QUALIFY -> biosafety, data, scale, affordability and status

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Trace discovery, validation, scale-up,...